

RESEARCH INTERESTS

My research goal is to develop assistive and rehabilitation robots that bridge the gap between humans and machines, enabling robotic systems to function as intuitive extensions and partners in human activity. My interests span both physical and cognitive human–robot interaction, including human movement analysis, upper-limb exoskeletons, musculoskeletal modeling, human-in-the-loop control, and social robotics.

EDUCATION

University of Pennsylvania

Philadelphia, PA — May 2026

- M.S.E. in Robotics (submatriculation), GPA: 4.0/4.0
- Master’s Thesis: *Musculoskeletal Digital Twin Model for Real-Time Human-in-the-Loop Feedback Control Systems*
Thesis Advisor: Dr. Nadia Figueroa
- B.S.E. in Bioengineering, GPA: 4.0/4.0

RESEARCH EXPERIENCE

GRASP Laboratory, University of Pennsylvania

Aug 2023 – Present

Graduate Research Assistant → Research Engineer

Philadelphia, PA

Hybrid Simulation- and EMG-Controlled Upper-Limb Exoskeleton

Jan 2023 – Present

- Developed a MyoSuite musculoskeletal digital twin using inverse dynamics, PD control, and quadratic programming, reproducing measured elbow kinematics with a mean tracking RMSE of 1.866°.
- Developed a causal temporal convolutional network that predicted subject-specific biceps and triceps EMG for unseen controller configurations with mean NRMSEs of 10.7% and 10.5%, respectively, at real-time-capable latency.
- Integrated surface EMG, OptiTrack motion capture, motor torque, and exoskeleton encoder data across three participants, four external loads, and five controller configurations.

EMG-Based Achilles Tendinopathy Classification

Aug 2023 – Present

- Developed a real-time high-density EMG collection pipeline using custom 3D-printed hardware and laser-cut MMX/silicon fabric electrodes for movement and injury analysis.
- Built a 3D variational autoencoder to quantify Achilles tendinopathy severity, assess injury laterality, and classify rehabilitation exercises from high-density EMG.
- Detected Achilles tendinopathy in an average of 20 of 21 held-out evaluations and correctly classified healthy, unilateral, and bilateral pathology in 18 of 21.

PROFESSIONAL EXPERIENCE

Research Resident

Jul 2026 – Present

Maingen (Y Combinator)

Remote

- Research machine-learning methods for generating and optimizing robotic end-effector CAD geometries using simulation-derived performance requirements.
- Develop simulation and evaluation pipelines for assessing generated designs through geometric validity, mechanical performance, and task-specific engineering constraints.

Founding Mechanical Engineer

Aug 2025 – Present

Tadashi Robotics

Philadelphia, PA

- Translate Ember from an academic social-robotics research platform into an early-stage product for rehabilitative patient support and naturalistic human–robot interaction.
- Designed the mechanical enclosure and embedded architecture, integrating wireless communication, audio I/O, vision, display, power, and motor-control hardware.
- Developed LLM-based interaction logic for reminiscence dialogue and emergency guidance and authored human-subject and non-human-subject IRB submissions supporting studies of Ember.

PUBLICATIONS

Maggie Wagner, Sanghyub Lee, **Nathaniel Kim**, Michael North, Nadia Figueroa, Josh Baxter, and Flavia Vitale. “Latent HD-sEMG Representation Learning for Achilles Tendinopathy Assessment.” Manuscript in preparation for submission to *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026.

Nathaniel Kim, Sanghyub Lee, Dhyey Shah, Jason Chen, and Nadia Figueroa. “Musculoskeletal Digital Twin Modeling for Biomechanical State Estimation During Weighted Elbow Flexion.” Under review at the *2026 IEEE-RAS International Conference on Humanoid Robots (Humanoids)*.

RESEARCH & DESIGN PROJECTS

- Verdigris: Assistive Ultrasound Guidance Device for Lumbar Puncture** Aug 2025 – May 2026
- Co-developed an assistive medical device that uses ultrasound-based bone detection and real-time angle feedback to guide clinicians toward viable lumbar puncture needle trajectories.
 - Designed the device enclosure and visual interface, integrated the ultrasound transducer, motor, needle-angle sensor, and control electronics, and conducted acoustic signal acquisition testing on tissue and spinal phantoms.
 - Validated angle sensing and LED feedback to within 1° across controlled targets; contributed to a benchtop prototype achieving 80% optimal needle-path detection and 95% bone-detection accuracy.
- Rehabilitative Driving Simulator** Jan 2023 – May 2023
- Collaborated with engineers and clinicians in Jamaica to develop a Unity-based driving simulator with custom steering and pedal controls for pediatric patients with hemiplegia.

PRESENTATIONS & POSTERS

- Penn Undergraduate Research Mentoring Program Symposium** 2024
Poster Presenter *University of Pennsylvania*

TEACHING & MENTORSHIP

- Tutor & Youth Mentor** Dec 2022 – Present
The Salvation Army *Philadelphia, PA*
- Founded and taught a 3D-printing course, tutored students in STEM subjects, and prepared students for competitive high-school entrance examinations.
 - Mentored students in exploring career pathways and interests in technology policy, computing, robotics, and game design.
- Teaching Assistant** Aug 2025 – May 2026
Bioengineering Modeling, Analysis and Design Laboratory I & II, University of Pennsylvania *Philadelphia, PA*
- Helped develop laboratory curriculum and instructional materials for MATLAB-based systems modeling and bioengineering design.
 - Supported laboratory instruction through technical debugging, individualized guidance, office hours, and exam-review sessions.
- Residential Advisor & STEM Mentor** May – Aug 2024; May – Aug 2025
BOLD Summers *Philadelphia, PA*
- Mentored high-school students through individualized research, college, and STEM-career guidance while designing and facilitating academic, professional-development, and community programming for more than 100 summer-program participants.

TECHNICAL SKILLS

- Programming & Tools:** Python, MATLAB, C#, MicroPython, ROS, MuJoCo, Unity, PyTorch, scikit-learn
- Modeling & Control:** Machine learning, musculoskeletal modeling, inverse dynamics, quadratic programming, feedback control, signal processing
- Experimental Systems:** Surface and high-density EMG, OptiTrack motion capture, acoustic signal acquisition, sensor integration, human-subject research
- Design & Fabrication:** SolidWorks, Onshape, Rhinoceros 3D, circuit design, 3D printing, laser cutting

HONORS & AWARDS

- Hugo Otto Wolf Memorial Prize, Penn Engineering** 2026
- Received a \$500 prize as the Bioengineering senior whose work demonstrated exceptional thoroughness and originality.
- Career Services Summer Funding Award** 2025
- Awarded \$5,000 in competitive funding to support full-time summer research.
- Penn Undergraduate Research Mentoring Program (PURM)** 2024
- Selected for a \$5,000 faculty-mentored undergraduate research award at the University of Pennsylvania.